A Metric Counterexample: Nearly Isometric Embeddability Need Not Imply Almost Lipschitz Embeddability

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Source Problem

This packet addresses Problem 4 in Section 5 of:

> Florent Baudier and Gilles Lancien, *Tight embeddability of proper and stable metric spaces*, arXiv:1503.06089.

The source problem, on PDF page 17, asks:

> Exhibit two metric spaces X and Y such that X nearly isometrically embeds into Y, but X does not almost Lipschitz embed into Y.

Definitions Used

We use the definitions from the source paper. A metric space X almost Lipschitz embeds into Y if there are constants $r > 0$, $D \geq 1$, such that for every continuous $\varphi : [0, \infty) \to [0, 1)$ with $\varphi(0) = 0$ and $\varphi(t) > 0$ for $t > 0$, there is a map $f_\varphi : X \to Y$ satisfying

$$r\, d_X(x, y)\varphi(d_X(x, y)) \leq d_Y(f_\varphi(x), f_\varphi(y)) \leq Dr\, d_X(x, y).$$

For nearly isometric embeddability, the admissible compressions $\rho \in \mathcal{P}$ are continuous functions with $\rho(t) = t$ on $[0, 1]$, $\rho(t) \leq t$ for $t \geq 1$, and $\rho(t)/t \to 0$. The admissible expansions $\omega \in \Omega$ satisfy $\omega(0) = 0$, $t \leq \omega(t)$ on $[0, 1]$, $\omega(t) = t$ for $t \geq 1$, and $\omega(t)/t \to \infty$ as $t \downarrow 0$. The space X nearly isometrically embeds into Y if for every such pair (ρ, ω) there is $f : X \to Y$ with

$$\rho(d_X(x, y)) \leq d_Y(f(x), f(y)) \leq \omega(d_X(x, y)).$$

Proof Intuition

The source definition of nearly isometric embeddability permits the embedding to depend on the requested sublinear compression ρ. We exploit this by building a target Y with one ray for each admissible ρ. The ρ-ray is metrically distorted by a sublinear function g_ρ that still dominates ρ, so the usual ray $\mathbb{N}_0$ can satisfy that particular nearly isometric request by entering that component.

The obstruction to almost Lipschitz embeddability is global. An almost Lipschitz embedding of the integer ray would become a bi-Lipschitz embedding after choosing a test function φ bounded below on all integer distances. But the bouquet target has no bi-Lipschitz ray. A Lipschitz image with linear lower growth must eventually be far from the common basepoint; at that height it cannot switch components, because switching costs the sum of the two heights. Once trapped in one component, distances grow only sublinearly, contradicting linear compression.

Auxiliary Concave Majorants

Lemma 1. *Let $\rho : [0,\infty) \to [0,\infty)$ be continuous, satisfy $\rho(t) = t$ on $[0,1]$, $\rho(t) \le t$ for $t \ge 1$, and $\rho(t)/t \to 0$ as $t \to \infty$. Then there is a continuous, nondecreasing, concave, unbounded function $g : [0,\infty) \to [0,\infty)$ such that*

$$g(0) = 0, \qquad g(t) = t \ (0 \le t \le 1), \qquad \rho(t) \le g(t) \le t, \qquad \frac{g(t)}{t} \to 0.$$

Consequently $d_g(m,n) = g(|m-n|)$ is a metric on $\mathbb{N}_0$.

Proof. Let $h(t) = t$ for $0 \le t \le 1$, and $h(t) = 1+\log t$ for $t \ge 1$. Then h is continuous, nondecreasing, concave, unbounded, $h(t) \le t$, and $h(t)/t \to 0$. Put $\rho_0(t) = \max\{\rho(t), h(t)\}$. Then $\rho_0(t) \le t$, $\rho_0(t) = t$ on $[0,1]$, and $\rho_0(t)/t \to 0$.

Choose a sequence $\varepsilon_j \downarrow 0$. Since $\rho_0(t)/t \to 0$, for each j the number

$$A_j = \sup_{t \ge 0} \bigl(\rho_0(t) - \varepsilon_j t\bigr)$$

is finite. Define

$$g(t) = \inf\Bigl(\{t\} \cup \{A_j + \varepsilon_j t : j \ge 1\}\Bigr).$$

Every affine function in the infimum dominates ρ_0, and the function $t \mapsto t$ also dominates ρ_0. Hence $g \ge \rho_0 \ge \rho$. Also $g \le t$. Because $g \le t$ and $g \ge \rho_0 = t$ on $[0,1]$, we have $g(t) = t$ on $[0,1]$.

The pointwise infimum of increasing affine functions is concave and nondecreasing. For each fixed j,

$$\limsup_{t\to\infty} \frac{g(t)}{t} \le \limsup_{t\to\infty} \frac{A_j + \varepsilon_j t}{t} = \varepsilon_j,$$

and therefore $g(t)/t \to 0$. Finally, $g \ge h$, so g is unbounded.

A nondecreasing concave function with $g(0) = 0$ is subadditive: concavity implies $g(s)/s \ge g(s+t)/(s+t)$ and $g(t)/t \ge g(s+t)/(s+t)$ for $s,t > 0$, whence $g(s) + g(t) \ge g(s+t)$. Thus $g(|m-n|)$ satisfies the triangle inequality on $\mathbb{N}_0$. □

The Counterexample

Theorem 1. *There are metric spaces X and Y such that X nearly isometrically embeds into Y, but X does not almost Lipschitz embed into Y.*

Proof. Let $X = \mathbb{N}_0$ with its usual metric $d_X(m,n) = |m-n|$. For every admissible $\rho \in \mathcal{P}$, choose a function g_ρ supplied by Lemma 1. Let $Y_\rho = \mathbb{N}_0$ with metric

$$d_\rho(m,n) = g_\rho(|m-n|).$$

Now form the metric wedge Y of the family $(Y_\rho)_{\rho\in\mathcal{P}}$ by identifying all zero points as a common basepoint o. If $a \in Y_\rho$ and $b \in Y_\sigma$, set

$$d_Y(a,b) = \begin{cases} g_\rho(|a-b|), & \rho = \sigma, \\ g_\rho(a) + g_\sigma(b), & \rho \ne \sigma. \end{cases}$$

This is the usual wedge metric: inside each component we use d_ρ, and between different components we travel through o.

We first show that X nearly isometrically embeds into Y. Fix $\rho \in \mathcal{P}$ and $\omega \in \Omega$. Define $F_{\rho,\omega} : X \to Y$ by $F_{\rho,\omega}(n) = n$ in the ρ-component. If $m, n \in \mathbb{N}_0$ and $k = |m-n|$, then for $k = 0$ there is nothing to check. For $k \geq 1$, the defining properties of g_ρ and the fact that $\omega(k) = k$ give

$$\rho(k) \leq g_\rho(k) = d_Y(F_{\rho,\omega}(m), F_{\rho,\omega}(n)) \leq k = \omega(k).$$

Thus X nearly isometrically embeds into Y.

It remains to show that X does not almost Lipschitz embed into Y. We first prove that Y contains no bi-Lipschitz copy of the usual ray. Suppose, toward a contradiction, that $F : \mathbb{N}_0 \to Y$ and constants $c, C > 0$ satisfy

$$c|m-n| \leq d_Y(F(m), F(n)) \leq C|m-n| \qquad (m, n \in \mathbb{N}_0).$$

Let $R_n = d_Y(F(n), o)$. Since

$$R_n \geq d_Y(F(n), F(0)) - R_0 \geq cn - R_0,$$

the heights R_n grow at least linearly. On the other hand, $d_Y(F(n+1), F(n)) \leq C$. If $F(n)$ and $F(n+1)$ lie in different components, then

$$d_Y(F(n+1), F(n)) = R_{n+1} + R_n,$$

which is eventually larger than C. Hence, for all sufficiently large n, consecutive points $F(n)$ and $F(n+1)$ lie in the same component. The tail of the sequence $F(n)$ is therefore contained in a single component Y_{ρ_0}.

Write $F(n) = a_n \in Y_{\rho_0}$ for all $n \geq N$. Since

$$g_{\rho_0}(|a_{n+1} - a_n|) = d_Y(F(n+1), F(n)) \leq C$$

and g_{ρ_0} is nondecreasing and unbounded, there is $B < \infty$ such that $|a_{n+1} - a_n| \leq B$ for all $n \geq N$. Therefore

$$|a_m - a_n| \leq B|m-n| \qquad (m, n \geq N).$$

For $m, n \geq N$,

$$d_Y(F(m), F(n)) = g_{\rho_0}(|a_m - a_n|) \leq g_{\rho_0}(B|m-n|).$$

Dividing by $|m-n|$ and letting $|m-n| \to \infty$, the right-hand side has ratio tending to 0 because $g_{\rho_0}(t)/t \to 0$. This contradicts the lower bound $d_Y(F(m), F(n)) \geq c|m-n|$. Thus no bi-Lipschitz embedding $\mathbb{N}_0 \to Y$ exists.

Finally suppose that X almost Lipschitz embedded into Y. Take

$$\varphi(t) = \begin{cases} t/2, & 0 \leq t \leq 1, \\ 1/2, & t \geq 1. \end{cases}$$

This φ is admissible in the almost Lipschitz definition. Since all nonzero distances in $X = \mathbb{N}_0$ are at least 1, the corresponding map would satisfy

$$\frac{r}{2}|m-n| \leq d_Y(f_\varphi(m), f_\varphi(n)) \leq Dr|m-n| \qquad (m \neq n),$$

which is a bi-Lipschitz embedding of the ray into Y. This contradicts the previous paragraph. Hence X does not almost Lipschitz embed into Y. $\square$

Verification Notes

The construction is intentionally nonseparable in general, since it uses one component for every admissible compression function ρ. The source problem only asks for metric spaces, so no separability or Banach structure is being claimed. The proof uses no external theorem beyond elementary facts about concave metric transforms and wedge metrics.

Bounded Novelty Check

On 2026-06-15, the run indexes were searched for `1503.06089`, `nearly isometric`, and `almost Lipschitz`; no duplicate packet or attempt for this metric problem was found. Local parsed-source search for phrases such as `nearly isometrically embeds` and `does not almost Lipschitz` found the source paper but no separate answer. Web searches for exact phrases including `"nearly isometrically embeds" "almost Lipschitz"`, `"Exhibit two metric spaces" "nearly isometrically" "almost Lipschitz"`, and `site:arxiv.org "almost Lipschitz" "nearly isometric"` returned the source paper and no later answer. The novelty claim is therefore bounded by these searches.

References

- F. Baudier and G. Lancien, *Tight embeddability of proper and stable metric spaces*, arXiv:1503.06089.